Task 1:

1) Type and save the following programs in visual studio. Run these programs and observe their output.

```
#include <iostream>
using namespace std;
int main () {
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter='B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output:

```
A
B
```

2)

```
# include <iostream>
using namespace std;
int main ()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
```

```
    letter << 66;
    cout << letter << '\n';
    cin >> letter ;
    return 0;
}
```

output:
65
66

3)
```
# include <iostream>
using namespace std;
int main ()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B' ;
    cout << letter << '\n';
    cin >> letter ;
    return 0;
}
```

output:
65
66

Task 2

Take two integers as input from the user apply arthematic operations on them (+, −, *, /) as print them on screen.

```
#include <iostream>
using namespace std;
int main()
{
    int p=15;
    int q = 9;
cout << p+q<<endl;
cout<< p-q<< end;
cout << p*q<<endl;
cout << P/q << endl;
return 0;
}
```

output:

```
24
6
135
1
```

Task 3

Write the output of following code.

```
#include <iostream>
using namespace std;
int main ( )
```

```
{
    int val 1;
    int val 2;
    int val 3;
    cout << "please enter your 3 numbers:";
    cin >> val 1 >> val 2 >> val 3;
    cout << val 1 << "\n" << val 2 << "\n" <<
                                val 3 << "\n";
    cout << val 3 << "\n" << val 2 << "\n" <<
    system ("Pause"); val 1 "\n";
    return 0;
}
```

Output:

```
Please enter your 3 numbers:
2, 6, 8
2
6
8
8
6
2
```

Task 4

Write a C++ Program to calcutate distance between two points.

Note: x_1, y_1, x_2, y_2 are all double values.

Farimula: $d = (x_2 - x_1)^2 + (y_2 - y_1)^2$

```
#include <iostream>
#include <cmath>
using namespace std;
    int main()
{
    double x1, y1, x2, y2;
    double distance;
    cout << "Enter x 1:";
```

```
    cin >> x1;
cout << "Enter y1:";
    cin >> y1;
cout << "Enter x2:";
    cin >> x2;
cout << "Enter y2:";
    cin >> y2;
distance= sqrt(Pow(x2-x1), 2)+Pow(y2-y1),2));
    cout << "Distance between two points is
         " << distance << endl;
    return;
}
```

Output:

```
Enter x1: 1
Enter y1: 6
Enter x2: 4
Enter y2: 8
Distance between two points is 3.60555.
```

Task 5

2) Calculate the Area of a circle(area = Pi * r2)

```
#include <iostream>
using namespace std;
int main ()
{
    double r,      area;
const double pie =3.14159;
cout << "Enter radius of circle:";
    cin >> r;
area = pi * r * r;
```

```
    cout << "Area of a circle:" << area << endl;
    return 0;
}
```

Output:

Enter radius of circle: 7

Area of a circle: 153.938

---

2) Calcute the Area of a Rectangle(area = lenght * width).

```
# include <iostream>
using namespace std;
int main ()
{
    float lenght, width, area;
    cout << "Enter lenght of rectangle:";
    cin >> lenght;
    cout << "Enter width of rectangle:";
    cin >> width;
area= lenght * width;
cout << "Area of rectangle:" << area <<
                                endl;
    return;
}
```

Output:

Enter Lenght of rectangle: 8

Enter width of rectangle: 6

Area of rectangle: 48.

3) Calculate the area of a triangle
(area = base * height * .5)

```
#include <iostream>
using namespace std;
int main ()
{
    float base, height, Area;
    cout << "Enter base of triangle:";
    cin >> base;
    cout << "Enter height of triangle:";
    cin >> height;
    area = base * height * 0.5;
    cout << "Area of triangle:" << area <<
    endl;
    return 0;
}
```

Output:

```
Enter base of triangle: 5
Enter height of triangle: 9
Area of triangle: 22.5
```

Task 6

Total purpose A customer in a store is purchasing five items. The prices of the five items are:

- Price of item 1 = $12.95
- Price of item 2 = $24.95

- Price of item 3 = $6.95
- Price of item 4 = $14.95
- Price of item 5 = $ 3.95

Write a program that holds the prices of five items in five variables. Display each items price, the subtotal of the sale, the amount of sale tax, and the total. Assume the sales tax is 6%.

```
#include <iostream>
using namespace std;
int main()
{
   float item1 = 12.95f;
   float item2 = 24.95f;
   float item3 = 6.95f;
   float item4 = 14.95f;
   float item5 = 3.95f;
float subtotal, salestax, total;
const float Tax-Rate = 0.06f;
Subtotal = item1+item2+item3+item4+item5;
Salestax = subtotal + Tax-Rate;
  Total = subtotal + sales Tax;
cout << "price of item 1: $" item1 << endl;
cout << "price of item 2: $" << it-em2 << endl;
cout << "price of item 3: $" << item3 << endl;
cout << "price of item 4: $" << item4 << endl;
cout << "price of item 5: $" << item5 << endl:
```

```
cout << "subtotal of the sale : $" << subtotal <<
                                        endl;
cout << "Amount of sale tax (6%) : $" << sale
                                    tax << endl;
cout << "total : $" << total << endl;
return 0;
```

```
Price of item 1 : $12.95   <= Output
Price of item 2 : $24.95
Price of item 3 : $6.98
Price of item 4 : $14.95
Price of item 5 : $3.95
Subtotal of the sale : $63.75
Amount of sale-tax(6%) : $63.81
Total : $1.61149e-43
```

Bonus Task

Write a program that inputs a five-digit integer, separates the integer into its individual digits and prints the digits separated from one another by three spaces each.

```
#include <iostream>
using namespace std;
int main()
{
    int number;
    cout << "Enter a five-digit integer:";
```

```
cin>>number;
int digit1 = number /10000;
int digit 2 =(number/ 1000)% 10;
int digit 3 =(number/100)% 10;
int digit 4 =(number/ 10)% 10;
int digit 5 =( number% 10;
cout << digit1 << "  "
     << digit2 << "  "
     << digit3 << "  "
     << digit4 << "  "
     << digit5 << endl;
return 0;
}
```

Output:

```
Enter a five-digit integer: 10000
1  0  0  0  0
```

Home Task

1) Exchange write a program that needs in two variables and swaps the value of these variables. For example: if a variable 'var 1' contains 10 & variable 'var 2' contains 20, after the swap operation the variable 'var 1' contains 20 (value of var 2) and variable 'var 2' contains 10 (value of var 1.

```
#include <iostream>
using namespace std;
int main ()
{

   int var 1 = 30 , var 2 = 40, temp;
   //swapping logic
    temp = var1;
    var 1 = var 2;
    var2 = temp;
cout << "var 1:" << var 1 << endl;
cout << "var 2:" << var 2 << end;
   return 0;
}
```

output:

var 1 : 40

var 2 : 30

2) Make Upper write a program that prompts the user to enter an alphabet in small caps(a, b, c, ... z) and display the entered alphabet into its upper caps (A, B, C, ... Z).
hint ASCII.

```
#include <iostream>
using namespace std;
Int main ()
{
  char ch;
cout << "Enter a lowercase alphabet(a-z):";
  cin >> ch;
 cout << uppercase." << (char) toupper(ch)
<< endl;
  return;
}
```

Output:

Enter a lowercase alphabet(a-z): z

Upper case: Z

```cpp
#include <iostream>
using namespace std;
int main ( ) {
    char letter;
    letter='A';
    cout<<letter<<'\n';
    letter='B';
    cout <<letter <<'\n';
    cin>> letter;
    return 0;
}
```

A

B



```cpp
#include <iostream>
using namespace std;
int main ( ) {
    char letter;
    letter='65';
    cout<<letter<<'\n';
    letter='66';
    cout <<letter <<'\n';
    cin>> letter;
    return 0;
}
```

5
6

```cpp
#include <iostream>
using namespace std;
int main ( ) {
    int letter;
    letter='A';
    cout<<letter<<'\n';
    letter='B';
    cout <<letter <<'\n';
    cin>> letter;
    return 0;
}
```

65
66

```cpp
#include <iostream>
using namespace std;
int main ( ) {
    int p=15;
    int q=9;
    cout<<p+q<<endl;
    cout<<p-q<<endl;
    cout<<p*q<<endl;
    cout<<p/q<<endl;

}
```

```
24
6
135
1

[Program finished]
```

```
1  #include <iostream>
2  using namespace std;
3  int main ( ) {
4      int val1 ;
5      int val2 ;
6      int val3 ;
7      cout <<"please enter your 3 numbers:";
8      cin >> val1>> val2 >> val3;
9      cout << val1<< "\n" << val2 <<"\n" << val3<< "\
   n";
10     cout << val3 << "\n" <<val2 <<"\n" <<val1<<
   "\n";
11     system ("pause");
12     return 0;
13
14 }
```

```
please enter your 3 numbers:2 6 8
2
6
8 n8
6
2
Press any key to continue . . .
```

```cpp
#include <iostream>
using namespace std;
int main ( )
{
    float base, height, area ;
    cout<<"enter base of triangle:";
    cin>>base;
    cout<<"enter height of triangle:";
    cin>>height;
    area=base*height*0.5;
    cout<<"area of triangle:"<<area<<endl;
    return 0;

}
```

```
enter base of triangle:5
enter height of triangle:9
area of triangle:22.5

[Program finished]
```

```cpp
#include <iostream>
using namespace std;
int main ()
{

   float lenght, width, area;
   cout << "Enter lenght of rectangle:";
   cin >> lenght;
   cout << "Enter width of rectangle:";
   cin >> width;
   area = lenght * width;
   cout << "Area of rectangle:" << area << endl;

   return 0;
}

```

```
Enter lenght of rectangle:8
Enter width of rectangle:6
Area of rectangle:48

[Program finished]
```

```
1  #include <iostream>
2  using namespace std;
3  int main ( )
4  {
5      float item1=12.95f;
6      float item2=24.95f;
7      float item3=6.95f;
8      float item4=14.95f;
9      float item5=3.95f;
10     float subtotal, sale_tax, total;
11     const float tax_rate=0.06f;
12     subtotal=item1+item2+item3+item4+item5;
13     sale_tax=subtotal+tax_rate;
14     cout<<"price of item1:$"<<item1<<endl;
15     cout<<"price of item2:$"<<item2<<endl;
16     cout<<"price of item3:$"<<item3<<endl;
17     cout<<"price of item4:$"<<item4<<endl;
18     cout<<"price of item5:$"<<item5<<endl;
19     cout<<"subtotal of the
   sale:$"<<subtotal<<endl;
20     cout<<"amount of sale_
   tax(6%):$"<<sale_tax<<endl;
21     cout<<"total:$"<<total<<endl;
22         return 0;
23
24 }
```

```
price of item1:$12.95
price of item2:$24.95
price of item3:$6.95
price of item4:$14.95
price of item5:$3.95
subtotal of the sale:$63.75
amount of sale_ tax(6%):$63.81
total:$1.61149e-43

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int number;
    cout << "Enter a five-digit integer: ";
    cin >> number;

    int digit1 = number / 10000;
    int digit2 = (number / 1000) % 10;
    int digit3 = (number / 100) % 10;
    int digit4 = (number / 10) % 10;
    int digit5 = number % 10;

    cout << digit1 << " "
         << digit2 << " "
         << digit3 << " "
         << digit4 << " "
         << digit5 << endl;

    return 0;
}
```

```
Enter a five-digit integer: 10000
1   0   0   0   0

[Program finished]
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int var1 = 30, var2 = 40, temp;

    // Swapping logic
    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "var1: " << var1 << endl;
    cout << "var2: " << var2 << endl;

    return 0;
}
```

```
var1: 40
var2: 30

[Program finished]
```

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;

    cout << "Enter a lowercase alphabet (a-z): ";
    cin >> ch;

    cout << "Uppercase: " << (char)toupper(ch)
<< endl;

    return 0;
}
```

```
Enter a lowercase alphabet (a-z): z
Uppercase: Z

[Program finished]
```